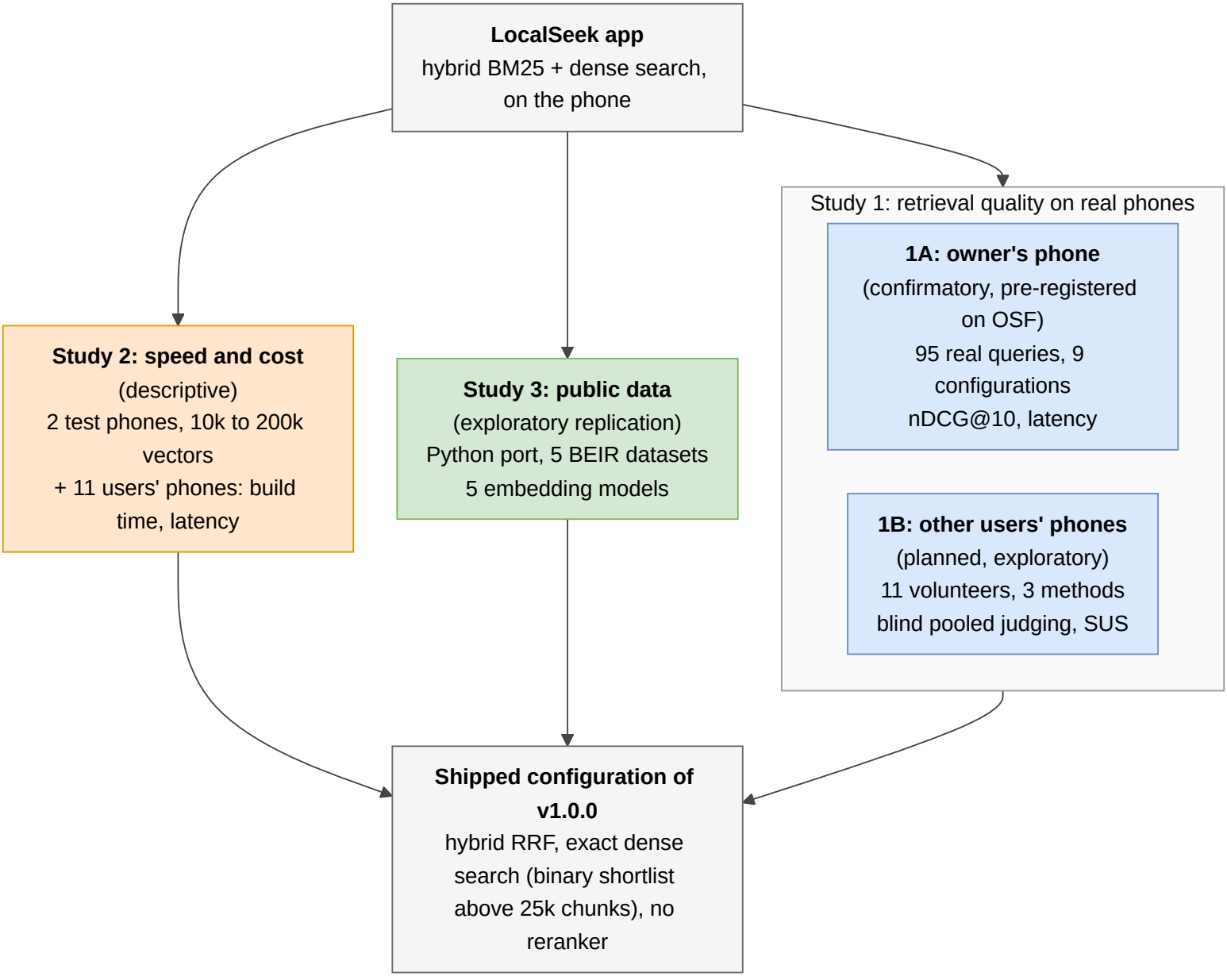
LocalSeek app
hybrid BM25 + dense search, on the phone
Study 1: retrieval quality on real phones
1A: owner's phone
(confirmatory, pre-registered on OSF)
95 real queries, 9 configurations
nDCG@10, latency
1B: other users' phones
(planned, exploratory)
11 volunteers, 3 methods
blind pooled judging, SUS
Study 2: speed and cost
(descriptive)
2 test phones, 10k to 200k vectors
+ 11 users' phones: build time, latency
Study 3: public data
(exploratory replication)
Python port, 5 BEIR datasets
5 embedding models
Shipped configuration of v1.0.0
hybrid RRF, exact dense search (binary shortlist above 25k chunks), no reranker